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Erasmus+ project ROBO STEAM ACADEMY
KA220-VET-7CF4F308

Lap Timer User and Setup Guide

Place, connect and check the version 2 enclosure · Version 2.0

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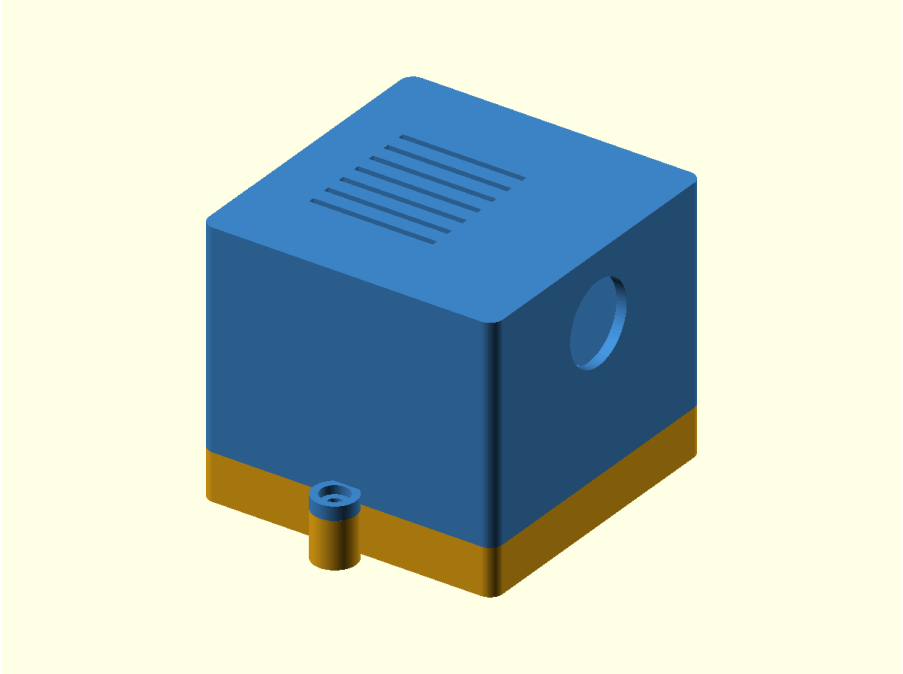
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About this guide

This guide covers the safe mechanical setup of the version 2 lap-timer enclosure at a robot course. The sensor axis is 45 mm above the supporting surface. The exact electrical wiring, firmware controls, platform protocol, and calibration indications must come from approved project sources when they are added.



Version 2 empty enclosure. The blue top and yellow bottom are separate printed parts.

1. Before each session

1. Disconnect power.
2. Check the enclosure for cracks or loose layers.
3. Check both closure lugs and screws.
4. Check that the sensor nut is secure.
5. Check that the cable is not cut, crushed, or sharply bent.
6. Check that the rear openings do not press on either plug.
7. Remove loose objects from the sensing path.

Check: the enclosure is complete, stable, and undamaged.

> **Safety**
>
> Do not use a cracked enclosure near moving robots. Replace or reprint the damaged part first.

2. Place the enclosure

1. Put the box on a flat surface beside the course.
2. Point the sensor horizontally across the robot path.
3. Keep the front face outside the robot's collision path.
4. Keep the rear connectors accessible.
5. Check that cables cannot pull the enclosure into the course.
6. Mark the tested position so it can be repeated.

Check: the box does not rock, the beam is horizontal, and a robot cannot hit the sensor face in normal operation.

2.1 Beam height

The version 2 optical axis is nominally 45 mm above the surface. The higher position gives the centred M18 body generous space above the estimated 24 mm electronics envelope.

Item	CAD value	Meaning
Optical-axis height	45 mm	Height of sensor centre above the table
Sensor body diameter	18 mm nominal	M18 threaded body
Nominal body bottom	36 mm	Axis minus 9 mm radius
Estimated electronics top	24 mm	Provisional design envelope
Nominal clearance	12 mm	Must be checked physically

3. Connect the equipment

Use only the approved circuit diagram, firmware instructions, power adapter, and competition-platform procedure. They are not yet present in this repository.

1. Keep all power disconnected.
2. Inspect the intended connector and cable.
3. Insert the micro-USB plug by holding its molded body.
4. Confirm that the plug seats fully without touching the enclosure wall.
5. Insert the barrel plug only after its voltage and polarity are approved.
6. Route both cables away from wheels and feet.

Check: each plug is fully seated, strain relief is not trapped, and no cable can pull the box into the course.

Safety

Never connect an unknown mains adapter. Check output voltage, polarity, current rating, connector fit, and local electrical-safety requirements.

4. Calibrate the sensor

Do not invent a sensitivity setting. Use the verified E18-D80NK and firmware procedure when it is added to the project.

1. Use the real robot and course surface.
2. Keep the box in its marked operating position.
3. Adjust only the control identified by the approved sensor instructions.
4. Move the robot through the detection area at realistic speed.
5. Repeat the test several times.
6. Record missed counts and false triggers.

Check: the same physical pass gives the expected system indication every time.

5. Run a session

1. Complete the pre-session inspection.
2. Place the enclosure on its marked position.
3. Connect the approved equipment.
4. Run the verified calibration check.
5. Keep people and loose objects out of the sensing path.
6. Observe the first robot passes.
7. Stop the session if the box moves or counting becomes unreliable.

Check: the enclosure stays in position and recorded events match observed passes.

6. End a session

1. Stop the timing system using its approved procedure.
2. Disconnect power.
3. Remove plugs by holding their molded bodies.
4. Inspect the sensor face and enclosure.
5. Record faults before moving the box.
6. Store the box where the sensor and lugs cannot be hit.

Check: the equipment is unpowered, clean, dry, and ready for inspection.

7. Version 2 fit report

After the first full session, record:

- whether the 45 mm beam height detects every intended robot;
- whether the box moves or tips;
- whether either rear opening limits plug insertion;
- whether the sensor or cable touches the electronics;
- whether the top can be removed without pulling wires;
- whether screw bosses remain sound after repeated access.

Think

If detection is reliable but the box moves, change the mounting or stability design. Do not change sensor geometry without evidence that beam height is the problem.

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